



Research Paper

Efficiency of chemically recuperated gas turbine fired with methane: Effect of operating parameters

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ABSTRACT

Previously, a set of optimistic studies have been published to show the efficiency of chemically recuperated gas turbines (CRGT) based on steam methane reforming. The thermodynamic efficiency of such turbines is depending not only on temperature and pressure in a cycle but also on temperature, pressure and steam-to-methane ratio in a recuperation system. In this paper, the effect of operational parameters (temperature, pressure, steam-to-methane ratio) on the efficiency of CRGT was determined. Exhaust heat recuperation in CRGT is taking place due to methane reforming, steam-methane mixture heating, and steam generation. A thermodynamic enthalpy of steam methane reforming process was determined via Gibbs free energy minimization method. The thermodynamic analysis of CRGT was performed in Aspen HYSYS. The efficiency of CRGT was determined for a wide range of operational parameters (gas turbine inlet temperature of 800–1500 °C, pressure of 5–25 bar, steam-to-methane ratio of 1...3). Based on the analysis it was confirmed that the efficiency of CRGT and thermochemical exhaust heat recuperation system is increasing with an increase in temperature. An increase in temperature and steam-to-methane ratio leads to an increase in a share of recuperated heat in a reformer. The heat recuperation rate is up to 0.63 for a high-temperature gas turbine and a steam-to-methane ratio of 3. The efficiency of CRGT is up to 49% for gas turbines with inlet gas temperature of 1500 °C which is an available level in modern gas turbines from Mitsubishi, Siemens, Alstom, etc.

1. Introduction

In recent years, there has been a steady trend in global energy towards a transition to renewable and carbon-neutral energy sources. However, the natural anomalies of 2020–2021, for example, in Europe and the United States, increased the demand for traditional energy sources — oil, natural gas, coal. However, among traditional fossil fuels, the “cleanest” fossil fuel is natural gas. That is why natural gas remains and will remain one of the main types of fuel in the coming decades. In addition, the IEA report “Golden Rules for a Golden Age of Gas” shows that natural gas (together with shale gas) will be the primary source of energy in the 21st century for the global economy [1].

Gas turbine plants for electricity generation are one of the biggest consumers of natural gas in the world [2,3]. Gas turbine units (GTU) have the following advantages: relatively low cost, high degree of automation of all processes, low impact on the environment, manoeuvrability and much more [4,5]. The relatively low cost of gas turbine units is ensured by the fact that the GTU scheme does not include such expensive elements of the steam turbine cycle as a steam boiler, a

condensing steam turbine unit with a service water supply system, and a water treatment unit.

Various engineering solutions can be applied to increase the efficiency of GTU: addition of a steam cycle [6], recuperation of exhaust heat by thermophysical recuperation (air heating) [7], steam injection to the combustion chamber (STIG) [8,9], etc. Thermochemical recuperation of exhaust heat can be considered as a relatively innovative way of energy efficiency improvement of gas turbines. The main concept of thermochemical exhaust heat recuperation is the use of exhaust heat after gas turbine for the endothermic process of steam methane reforming to produce hydrogen-rich synthesis gas [10,11]. This synthesis gas after the recuperation system is used as a fuel in GTU. Gas turbines with thermochemical exhaust heat recuperation were named chemically recuperated gas turbines.

Olmsted and Grimes were the first who proposed using thermochemical recuperation based on steam methane reforming for the gas turbines [12]. The authors suggested partially replacing of steam superheater section with a steam methane reformer. The exhaust heat after a gas turbine was used for the endothermic steam methane reforming process. Then the products of reforming are fed into the

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turbine combustor. In that paper, Olmsted and Grimes showed that $\text{CH}_4/\text{H}_2\text{O}$ absorbs exhaust heat thermally (as the temperature of mixture increases), and chemically (as the endothermic process occurs), resulting in a higher potential recuperation of exhaust heat than can be observed by conventional recuperation using only air heating. Further, the idea of using TCR in gas turbines was widely discussed in the scientific literature.

Various schemes of TCR implementation in the GTU were proposed in last years [13,14]. Sadeghi et al. [7] compared the performance of the organic Rankine cycle system and a combined gas turbine with thermophysical and thermochemical recuperation by steam methane reforming. It was found that the power generation system with thermophysical recuperators is the best from the exergy efficiency and energy generation perspective. However, a system with thermochemical recuperation had a lower sum of the unit cost of product value, which indicates a better economic performance.

Han et al. [15] studied a way to use chemical exergy of fuels prior to combustion in a power generation system based on coal and natural gas combustion. The authors proposed a two-step fuel transformation: the first step is coal gasification with air and steam and the second step is the combustion of residual char (about 40% of initial coal) to drive the steam-methane reforming reaction. After transformation fuel is mixed and combusted in a gas turbine, the residual heat of the flue gases is used in steam generation for the steam turbine. That technique led to an improvement in power generation by nearly 6%.

Carapellucci and Milazzo [16] considered implementation of steam methane reforming driven by exhaust gas heat in a reheat gas turbine. They established that efficiency of a gas turbine with thermochemical recuperation increases by 14.1% compared to a standard design. In a reheat gas turbine, there are also combustion chambers with high and low operating pressure. So if a steam methane reforming reactor is divided into two parts with various pressures, the process optimization improves gas turbine efficiency by an additional 1.4%.

Natural gas chemical transformation can also reduce the gas turbine CO_2 output due to the dry reforming process. Cao and Zheng [17] investigated the power generation system with high-temperature waste heat utilization by the means of dry reforming. After reforming, exhaust gases had lower temperatures, so their heat was used for ammonia absorption refrigeration. Results indicate that power generation efficiency was improved up to 48.9%, but the excess amount of generated cooling required to export. The proposed system is less efficient than that implemented in reheat gas turbines but is able to capture all the CO_2 thus reaching net-zero emissions.

For the gas turbine exhaust recuperation, other fuels could be used as the flue gas temperature varies significantly on the power production. Tola and Lonis [18] presented a new scheme with coupled methanol production and hydrogen generation by renewable sources. Methanol production aside from hydrogen requires heat and CO_2 that is captured from the turbine exhausts. The gas turbine is fed solely by chemically recuperated fuel and has an exhaust temperature of 1000 °C and a power of 650 kW. System efficiency was established at 0.23 but after scaling the power generation to 3.2 MW, efficiency can reach up to 0.6. Also, it is noticeable that the authors used the Aspen Plus simulation model for their calculations.

Moreover, the following fact has to be highlighted that it is not considered in the details in the present paper. In the chemically recuperated gas turbines, hydrogen-rich fuel (synthesis gas) is used as a fuel for heat generation in a combustion chamber. There is sufficient information in the literature that replacing an original hydrocarbon fuel with the products of its reforming leads to decreasing in NO_x emission [19,20]. Lui et al. presented the results of the investigation of methane reforming and combustion characteristics in chemically recuperated gas turbines [21]. The authors stated that TCR can be considered as ultralow NO_x emission technology.

The efficiency of thermochemical exhaust heat recuperation depends on the efficiency of the steam methane reforming process. The

operation parameters such as temperature, pressure, inlet steam-to-methane ratio play important roles for methane conversion, synthesis gas composition, enthalpy of the reforming process, and as a result, these parameters affect the efficiency of the thermochemical recuperation systems. Previously, Pashchenko presented the results of thermodynamic analysis of the thermochemical waste-heat recuperation system in a gas turbine [22]. The main goal of that paper was determination of effect of operational parameters on the efficiency of thermochemical recuperation system but the effect of these parameters on the turbine efficiency was not investigated. On the other hand, the operational parameters such as pressure and temperature of gas before a gas turbine are determining the GTU cycle efficiency. Therefore, the thermodynamic efficiency of chemically recuperated gas turbines is depending on the temperature before and after the gas turbine, pressure and steam-to-methane ratio on the thermochemical recuperation system. Research studies on thermodynamic analysis of CRGT [16,23] were mainly focused on studies of recuperation systems. However, there is no literature (according to the authors) on thermodynamic analysis of CRGT fired with methane in terms of effects of operation parameters (temperature (before and after gas turbine), pressure and feed composition in TCR system) on the efficiency of CRGT.

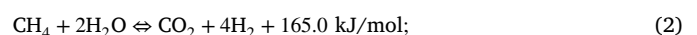
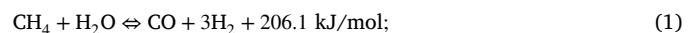
In the present paper, thermodynamic analysis of chemically recuperated gas turbine fired with methane was performed. The effect of operating parameters such as temperature (before and after gas turbine), pressure and feed composition in the TCR system on the gas turbine performances were studied. The thermochemical exhaust heat recuperation system based on steam methane reforming was analysed through the total Gibbs free energy minimization method via Aspen-HYSYS. The efficiency of CRGT and the recuperation system was determined as a function of operational parameters.

2. Thermochemical recuperation concept

According to the second law, one of the main directions of increasing the GTU efficiency is to increase the temperature of the gases before the gas turbine. Mitsubishi corporation developed J-series gas turbines with the inlet temperature capacity of 1600 °C [24]. SGT-series gas turbines produced by Siemens work with the inlet temperature up to 1500 °C. Gas turbines of G-series produced by Westinghouse have the inlet temperature of 1450 °C [25]. Other gas turbine manufacturers such as Alstom, General Electric, Skoda also have gas turbines with temperatures above 1500 °C. The high temperature of the gases before the gas turbine results in a high temperature of the gases after the gas turbine. Increasing the compression ratio decreases the temperature of the exhaust gases. However, increasing the compression ratio above the optimal one leads to a decrease in GTU efficiency. Therefore, for the above mentioned turbines, the temperature of the exhaust gases can reach 600–800 °C.

In this paper, we considered the gas turbines with thermochemical exhaust heat recuperation based on steam methane reforming. The efficiency of the steam methane reforming process strongly depends on temperature [26]. But in the temperature range above 600–700 °C the methane conversion leads to unity and the enthalpy of the process leads to maximum value. Therefore, it can be assumed that the exhaust gas temperature will provide the temperature potential for an efficient process of steam methane reforming.

The main concept of thermochemical exhaust heat recuperation is the use of hot exhaust gas after a GTU for a thermochemical transformation of initial fuel, for example, methane via steam reforming [27]. The steam methane reforming process is well studied and chemically this process can be described by three reactions:



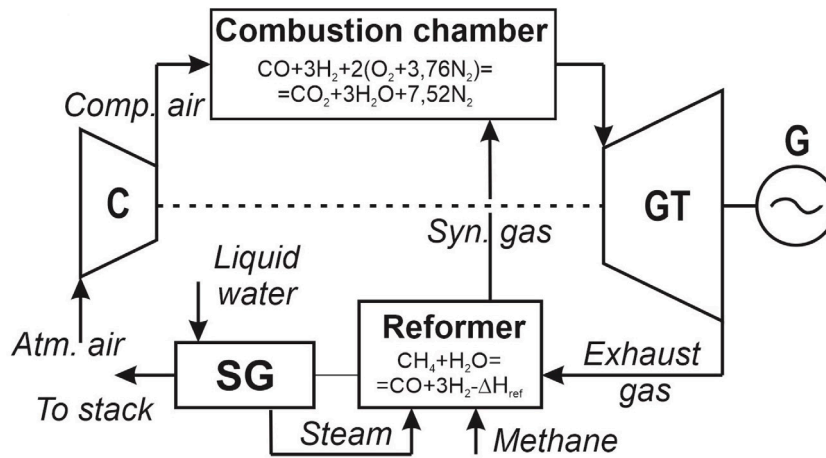
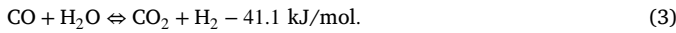


Fig. 1. The schematic diagram of heat generation from methane in STIG (a) and CRGT (b): C — compressor; GT — gas turbine; SG — steam generator; G — generator.



The thermochemical recuperation occurs in a methane reformer. In the methane reformer, the mixture of methane and steam is converted to the new fuel. Further, this initially transformed fuel (synthesis gas or syngas) is used in a combustion chamber. Since the reaction of steam methane reforming is strongly endothermic, external heat has to be supplied. In the thermochemical recuperation systems, the source of external heat is exhaust heat after the gas turbine.

The concept of thermochemical exhaust heat recuperation in GTU can be explained by the schematic diagram. Fig. 1 shows the schematic diagram of the chemically recuperated gas turbine fired with methane. As it can be seen from Fig. 1, in CRGT the heat extraction from methane is divided into two stages. The first stage is a thermochemical transformation of initial methane by steam reforming reaction. In this stage, new synthetic fuel with combustion heat higher than combustion heat of initial methane (in comparison to 1 kg of initial methane) is obtained. And the second stage is the use of this new synthetic fuel for heat generation by combustion in a combustion chamber.

In CRGT, exhaust heat is used for steam generation and for steam methane reforming reaction. The schematic diagram in Fig. 1 can be slightly expanded by the addition of a preheater of the steam-methane mixture but in the general case the steam-methane mixture can be preheated in the reformer. Therefore, in CRGT the exhaust heat is recuperated in the steam generator and reformer.

3. Methodology

One of the main criteria of the thermochemical recuperation system efficiency is a heat recovery rate (HRR). HRR is a relation between the recuperated heat and the enthalpy of exhaust heat after a gas turbine. The recuperated heat for CRGT is a sum of heat that has been transferred from exhaust gas to steam-methane mixture in the reformer for the reforming process and the mixture heating as well as heat for steam generation. In other words, the recuperated heat in CRGT can be determined as a sum of the enthalpy of the steam methane reforming process, change of the enthalpy of the steam-methane mixture by heating, and steam generation in the steam generator. Therefore, the heat recuperation rate can be determined via the following expression:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{mix} + \Delta H_{sg}}{H_{exh}} \quad (4)$$

where H_{rec} — a recuperated heat in the TCR system; H_{exh} — an exhaust gas enthalpy; ΔH_{298} — an enthalpy of the steam methane reforming

process; ΔH_{mix} — the enthalpy change of the steam-methane mixture in a reformer; ΔH_{sg} — the enthalpy for steam generation.

Moreover, in this paper, the recuperated heat for steam generation, steam-methane mixture preheating, and reforming process as a share of the total recuperated heat in TCR system was determined from the following expression:

$$r_{rec,i} = \frac{H_{rec,i}}{H_{rec,tot}} \quad (5)$$

where $H_{rec,i}$ is the recuperated heat in the i th element of the scheme (steam generator, reformer); $H_{rec,tot}$ is the total recuperated heat in the whole TCR system.

The enthalpy of the steam methane reforming process in the expression (4) is depending on the operational parameters such as temperature, inlet steam-to-methane ratio (mole of steam to mole of methane), and pressure. Therefore, the thermodynamic analysis includes not only the cycle thermodynamics but also the thermodynamic of the steam methane reforming process.

3.1. Thermodynamic cycle analysis

The thermodynamic model has been developed via Aspen HYSYS which is a powerful process simulation tool for the schematic design of various concepts. Based on the developed model, parametric analysis was performed for each case of CRGT. The main goal of this analysis was to determine which operating parameters have a significant effect on overall GTU efficiency. There is a large number of publications on using Aspen HYSYS for modelling of various schemes of GTUs [28,29].

The calculation scheme in Aspen HYSYS is presented in Fig. 2. In this schematic diagram, we added a preheater of the steam-methane mixture. The temperature of working fluid at the gas turbine inlet has been regulated by the air mass flow rate. Aspen HYSYS has a useful function, (*Adjust*), in which the user sets the target variable value and specifies an adjusted variable used to reach the target value. The temperature of exhaust gas has been set as the target variable and the air mass flow as the adjusted variable. Also, we used *Adjust* to compile the heat balance for the determination of exhaust gas temperature after a reformer, a heater, and a steam generator.

In this paper, the net work output of the gas turbine has been calculated based on the following expression:

$$W_{GTU} = W_{gt} - W_{comp} \quad (6)$$

where W_{gt} is the power produced in gas turbine; W_{comp} is the work for air compressing (compressor work).

The initial conditions and turbine parameters for a thermodynamic analysis are presented in Table 1.

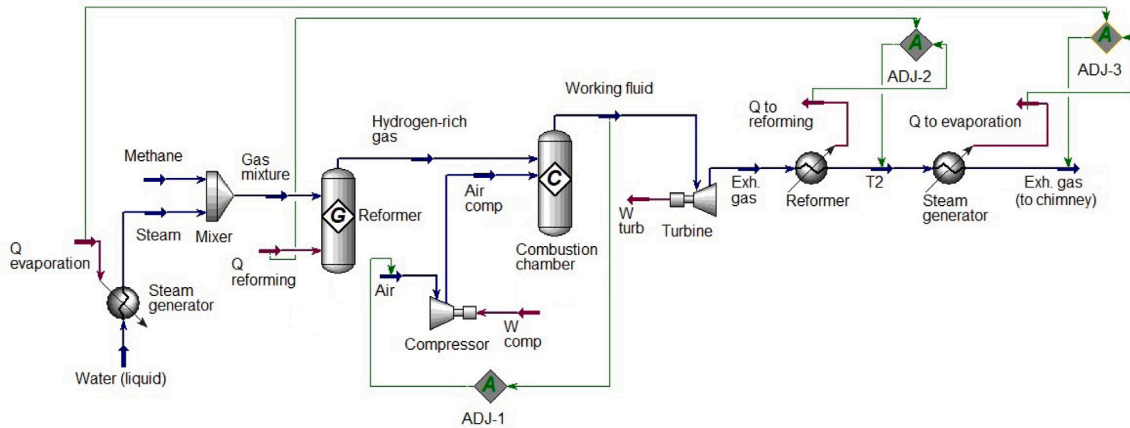


Fig. 2. The computational scheme of methane fired chemically recuperated gas turbine in Aspen HYSYS.

Table 1

Stream properties and efficiency parameters of gas turbine and compressor.

Parameter	Value
CH ₄ mass flow rate	100 kg/h
Gas temperature at GT inlet (T_{in})	800–1500 °C
Compression/expansion ratio	5; 10; 15; 20; 25
Pressure losses	0
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Steam temperature	Saturated
Steam-to-methane ratio (mol-to-mol)	1.0; 2.0; 3.0
Inlet temperature of water	20 °C
Temperature of atmospheric air	20 °C
Pressure of all inlet streams (except air)	10 bar

3.2. Thermodynamic analysis of SMR

The enthalpy of the steam methane reforming (SMR) process in the expression (4) is depending on the operational parameters such as temperature, inlet steam-to-methane ratio, and pressure. In addition, these operational parameters affect the composition of synthesis gas after a reformer. In the thermodynamic analysis of SMR, the equilibrium conditions are analysed. Therefore, the type of catalyst, reformer, etc have a negligible impact on the reforming products composition and the enthalpy of the process. To determine the SMR process enthalpy and synthesis gas composition as a function of the operational parameters, the thermodynamic analysis via Gibbs free energy minimization method was performed.

The thermodynamic analysis makes it possible to calculate the equilibrium concentration of the reforming products at the given operating parameters such as temperature, inlet steam-to-methane mixture and pressure. The main concept of the Gibbs free energy minimization method is the fact, that the chemical system is thermodynamically favourable when its total Gibbs free energy is minimum and its differential is equal to zero [30,31]:

$$dG^t = 0 \quad (7)$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the SMR decomposition process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \quad (8)$$

where n_i is number of i th component moles; μ_i is chemical potential of i th component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (9)$$

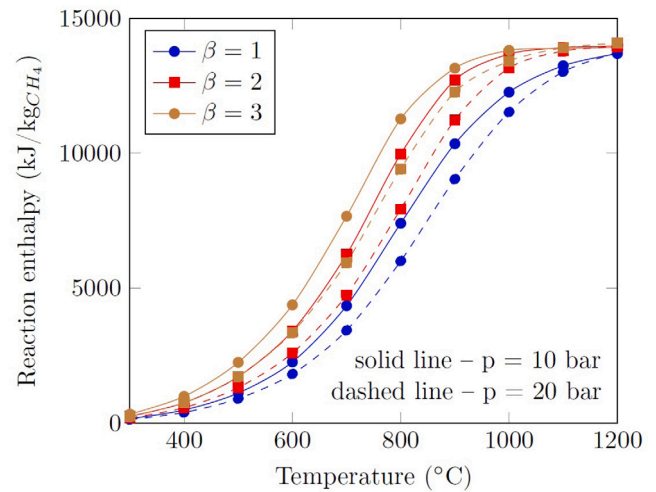


Fig. 3. The enthalpy of the steam methane reforming process as a function of a temperature at the steam-to-methane ratios (β) of 1, 2, 3.

where G_{fi}^0 is standard Gibbs free energy of i th component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure i th component; f_i^0 is fugacity of a pure i th component at the standard state.

If the equations presented above will be combined, the equation for total Gibbs free energy of the chemical system can be written as the following:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (10)$$

The thermodynamic analysis of SMR was performed also via Aspen HYSYS software that is a widely used program in chemical engineering for the modelling of chemical systems similar to the considered thermochemical recuperation system. To simulate the steam methane reforming process, the ideal Gibbs reactor is chosen. The Gibbs reactor determines the reforming products by the Gibbs free energy minimization method of each component based on the chemical mechanism of the SMR process. The Peng–Robinson equation of state was used to calculate the volume of the product gas as a function of pressure and temperature [32].

3.3. Validation

Although Aspen HYSYS is a generally recognized tool for modelling chemical processes, verification of the thermodynamic analysis results

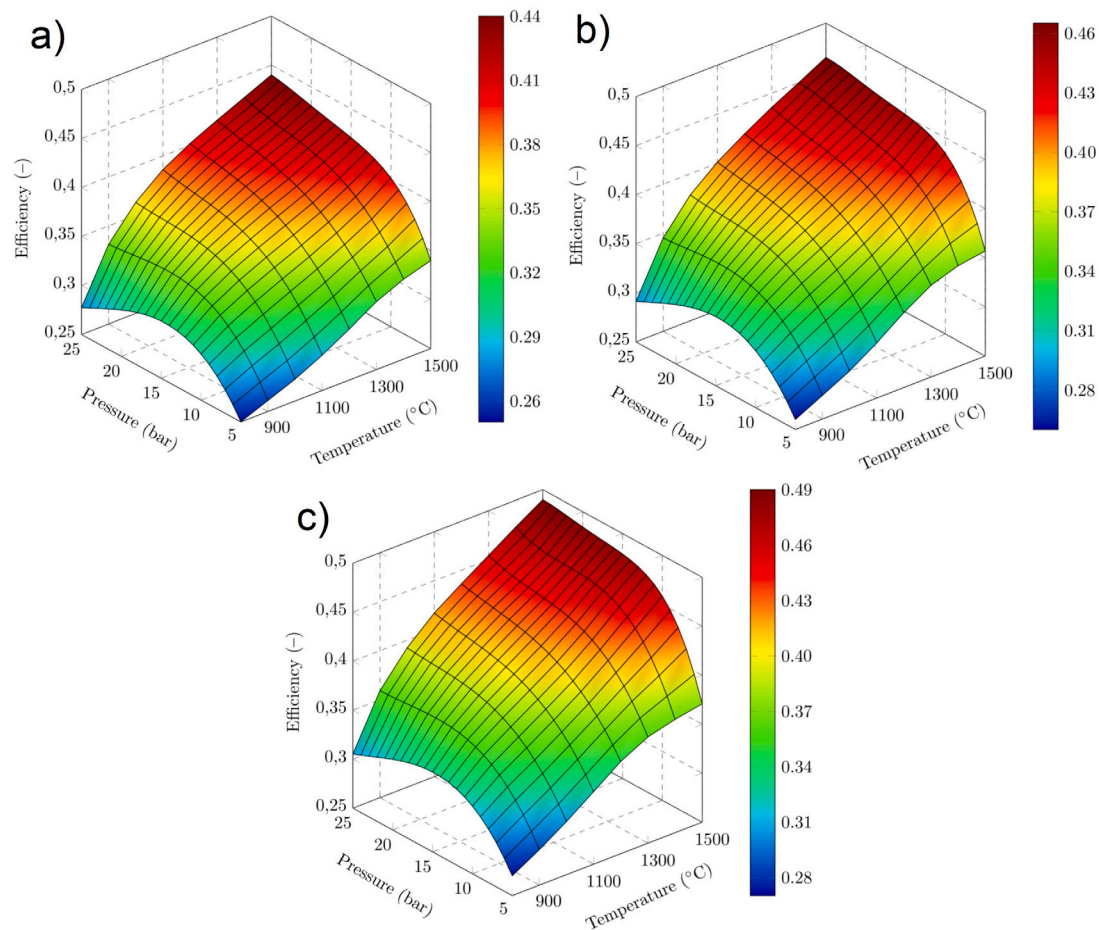


Fig. 4. GTU efficiency as a 3D-function of temperature at the turbine inlet (T_{in}) and pressure of a gas stream before gas turbine at the steam-to-methane ratios (β) of 1 (a), 2 (b), 3 (c).

of the SMR process was carried out in this article. The obtained results were compared with the results of Verkhivker and Karavchenko [33]. Moreover, the results were compared with the results of a classical thermodynamic analysis based on a conjugate solution of material balance and law action mass equations [26].

The thermodynamic analysis of the SMR process was carried out for various operating parameters. A comparison of the results of the synfuel composition is shown in Table 2. The upper results are calculated via Aspen HYSYS, the middle rows are the mole fraction from the paper presented by Verkhivker and Kravchenko [33], the bottom results are the mole fraction of each component that was calculated based on the classical thermodynamic analysis [26]. A good correlation is observed for all results.

Moreover, the comparison of the thermodynamic analysis results with the results of other authors [34,35] showed a good correlation for the wide range of the operational conditions.

Carbon formation is another factor that has to be discussed in the papers dealing with steam methane reforming, especially at a low steam-to-methane ratio at the reformer inlet. Carbon formation and deposition on the catalysts can be controlled by the composition of catalysts and by regulating the steam-to-methane ratio. There is sufficient information in the scientific papers that carbon deposition has a negligible effect at the steam-to-methane ratio above unity at a high carbon-resistance catalyst [36–38]. In addition, the previous authors' study showed that proper management in the steam methane reforming process avoids the carbon deposition [39].

Table 2

The comparison of the synthesis gas composition at a steam-to-methane ratio of 2, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000 K	1 bar	HYSYS	63.55	14.22	16.11	5.22	0.9
		Verkhivker [33]	63.42	14.43	16.11	5.12	0.92
		Classical [26]	63.01	14.32	16.32	5.23	1.12
1000 K	5 bar	HYSYS	61.08	14.85	17.57	5.13	1.55
		Verkhivker [33]	61.44	14.93	17.97	4.18	1.48
		Classical [26]	60.66	14.82	17.32	5.55	1.65
1100 K	1 bar	HYSYS	63.78	15.95	16.2	3.98	0.09
		Verkhivker [33]	63.76	16.11	16.11	3.94	0.08
		Classical [26]	63.88	16.08	16.02	3.93	0.09
1100 K	5 bar	HYSYS	61.24	14.71	18.11	4.28	1.66
		Verkhivker [33]	61.45	14.93	17.96	4.18	1.48
		Classical [26]	61.28	15.02	17.87	4.24	1.59

4. Results and discussion

4.1. Cycle analysis

The initial conditions and turbine parameters for the thermodynamic analysis are presented in Table 1.

According to the schematic diagram from Fig. 1, exhaust heat after the gas turbine is supplied to the reformer for the reforming process and heating the steam-methane mixture and to the steam generator

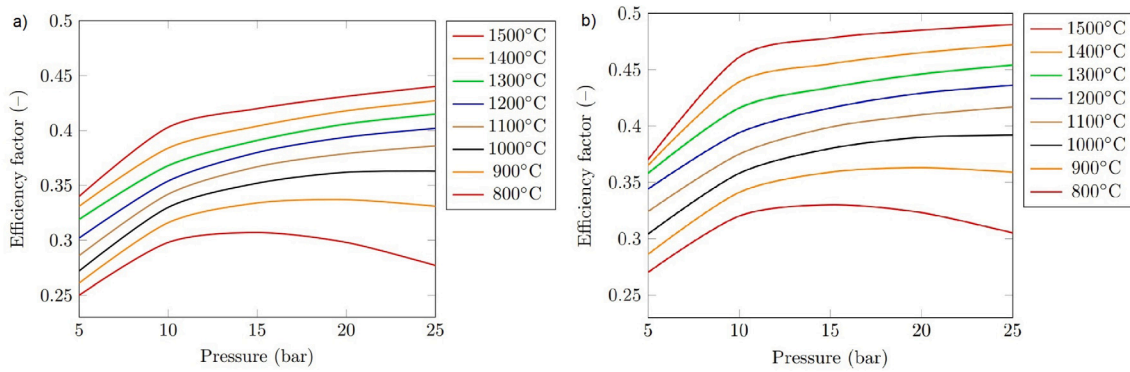


Fig. 5. GTU efficiency as a 2D-function of temperature (T_{in}) at the steam-to-methane ratios (β) of 1 (a), 3 (b) for various pressures at gas turbine inlet.

for the steam generation. The efficiency of CRGT depends on the efficiency of the recuperation system. In turn, the efficiency of the recuperation system significantly depends on the enthalpy of the steam methane reforming process. To understand the effect of temperature and steam-to-methane ratio on the enthalpy of the steam methane reforming process, a series of dependencies were obtained. Fig. 3 shows the enthalpy of the steam methane reforming process as a function of temperature at the steam-to-methane ratios (β) of 1, 2, 3 at a pressure of 10 bar.

Fig. 3 shows that in the temperature range before 600 °C the process enthalpy is slightly increasing. In the temperature range of 600–800 °C the process enthalpy is drastically increasing. And in the temperature range above 800 °C is reaching the maximum value of 13970 kJ/kg. Fig. 3 shows the effect of exhaust gas temperature and steam-to-methane ratio on the net work output of CRGT.

Fig. 3 shows that reaction enthalpy increases with an increase in a steam-to-methane ratio and decreases with increasing pressure. This fact is fully consistent with Le Chatelier's principle. Moreover, pressure and steam-to-methane ratio have the greatest effect on the reaction enthalpy in the temperature range from 600 to 1000 °C.

The main goal of the use of the thermochemical exhaust heat recuperation system is the increase of the gas turbine units efficiency by recuperation of exhaust heat back to the cycle.

To show the effect of operational parameters on the GTU efficiency, the GTU efficiency for various cases was calculated. The efficiency of the gas turbine unit was calculated based on the following expression:

$$\eta_{GTU} = \frac{W_{gt} - W_{comp}}{m_{CH_4} \cdot LHV_{CH_4}} \quad (11)$$

where m_{CH_4} is the mass flow rate of methane; LHV_{CH_4} is the methane lower heating value.

Fig. 4 shows the dependence of the GTU efficiency on temperature before gas turbine (T_{in}) and pressure of gas stream before gas turbine for different steam-to-methane ratios ($\beta = 1$ (a), $\beta = 2$ (b), $\beta = 3$ (c)). It can be seen from Fig. 4 that an increase in the β ratio leads to an increase in efficiency. An increase in the β ratio leads to an increase in steam consumption and, as a consequence, to an increase in the recuperated heat in the steam generator. A similar pattern can be seen in STIG turbines [40–42], where increasing steam injection increases the efficiency of the gas turbine. However, if the injection of steam in STIG leads mainly to a decrease in compressor work, in the CRGT the addition of steam also leads to an increase in the enthalpy of the reforming process and, as a consequence, to an increase in the recuperated heat in the reformer. To more clearly understand the effect of operational parameters on the GTU efficiency, the 2D-dependence of the efficiency vs. temperature for various β ratios and pressures is depicted in Fig. 5.

Figs. 4 and 5 show that an increase in the gas temperature at the turbine inlet increases the efficiency. This fact is fully consistent with

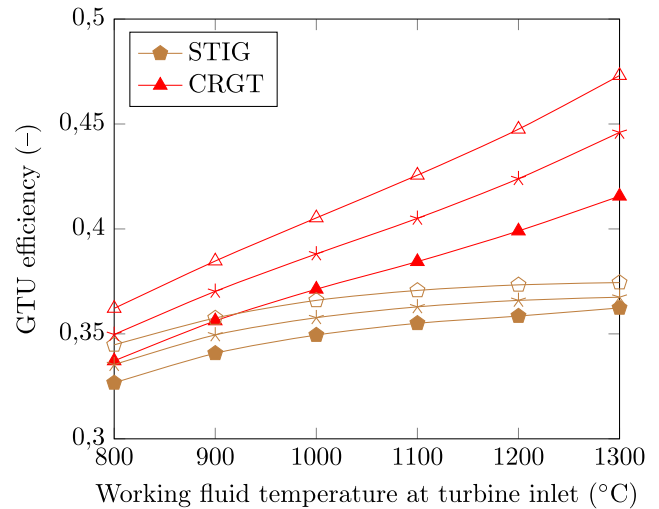


Fig. 6. The efficiency of STIG (a) and CRGT (b) as a function of temperature of working fluid at a gas turbine inlet for various steam mass flow rates: 112.5 kg/h (solid points); 225 kg/h (asterisk); 337.5 kg/h (hollow points).

the second law of thermodynamics. However, Figs. 4 and 5 clearly depict that increasing the pressure (compression ratio) above 10 bar does not significantly increase efficiency. In addition, for a gas temperature of up to 800–1000 °C, an increase in pressure above the optimal one leads to a decrease in efficiency. Such nature of the pressure influence can be explained by the following facts:

- in the temperature range above 1000 °C, an increase in pressure leads to an increase in the efficiency of the gas turbine cycle, but reduces the efficiency of the exhaust heat recuperation system because the enthalpy of the methane reforming process decreases with increasing pressure (according to Fig. 3);
- in the temperature range below 1000 °C, an increase in pressure above the optimum one leads to a decrease in the efficiency because in this case the temperature of the exhaust gases is insufficient for the implementation of the methane reforming process.

Fig. 4 shows that the efficiency of GTU with thermochemical recuperation can reach up to 0.5 (for $\beta = 3$) at a moderate (up to 25) compression ratio. It is expected that an increase in the steam-to-methane ratio above 3 will also increase the efficiency of the GTU. However, in this case, the synthesis gas will contain a large amount of incombustible steam, and at a moderately low temperature of the exhaust gases, there will be a heat deficit in the heat recuperation system. Exhaust heat will not be sufficient to generate steam and

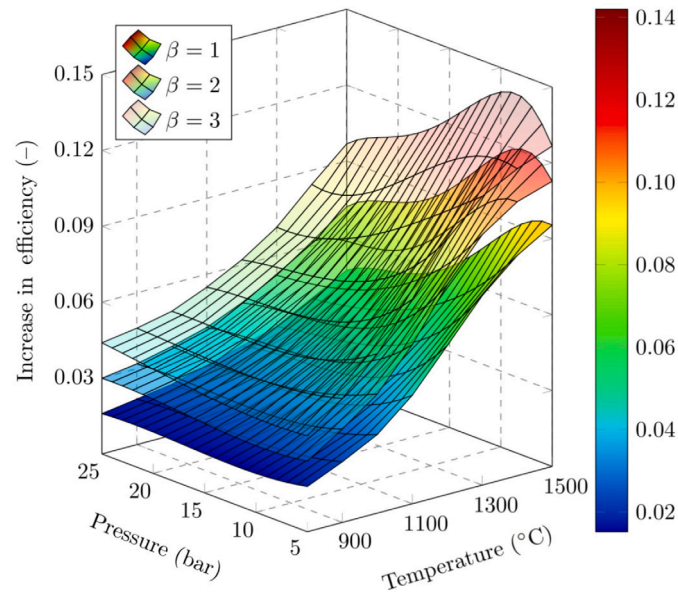


Fig. 7. Heat flow from exhaust gas after gas turbine for $\beta = 1, 2, 3$ and inlet pressure of 5 (solid) and 15 (transparent) bar.

heat the methane-steam mixture. Moreover, an increase in the steam-to-methane ratio leads to an increase in steam consumption and to an increase in the cost of water for its production. Therefore, the determination of the optimal steam-to-methane ratio is an important technical and economic task, which will be solved by the authors in the next studies.

To show the effect of adding of the thermochemical recuperation system in gas turbine on the efficiency, the comparison of two type turbines was performed: chemically recuperated gas turbine (CRGT) and steam injection gas turbine (STIG). Fig. 6 shows that CRGT is more efficient in comparison with STIG. The maximum efficiency has for the high-temperature gas turbines. The efficiency of STIG and CRGT at $T_{in} = 1300$ °C for a steam-to-methane ratio of 3 is 37.4% and 47.3%, respectively. the main reason for an increase in the efficiency of CRGT at $T_{in} = 800$ °C is preheating of methane as a part of a steam-methane mixture. With an increase in T_{in} the recuperated heat in a reformer is increasing. Therefore, a gas turbine with thermochemical exhaust heat recuperation is more efficient than STIG.

4.2. Recuperation system analysis

To assess the efficiency of using thermochemical exhaust heat recuperation, we compared the efficiency of the GTU unit without an exhaust heat recuperation system and GTU with a thermochemical exhaust heat recuperation (Fig. 7).

Fig. 7 shows that the increase in the efficiency of GTU due to thermochemical exhaust heat recuperation at low inlet gas temperature (T_{in}) is minimal. This can be explained by the fact that methane reforming is practically absent at a low temperature of exhaust gases and the heat recuperation system is looking like preheater (thermophysical recuperation). With an increase in temperature, the efficiency from the use of TCR increases. Fig. 7 shows that TCR allows increasing the efficiency of GTU up to 14.3%.

Fig. 4 makes it clearly that thermochemical exhaust heat recuperation is increasing the GTU efficiency. To understand the effect of operational parameters on the efficiency of the thermochemical exhaust heat recuperation system, we calculated the heat flows in a steam generator, a reformer and heat flow with exhaust gas to stack. Fig. 8 shows the heat flow from exhaust gas after a gas turbine to the reformer, steam generator, and to stack. In Fig. 8, the total column height is the heat flow with exhaust gas after a gas turbine. This heat

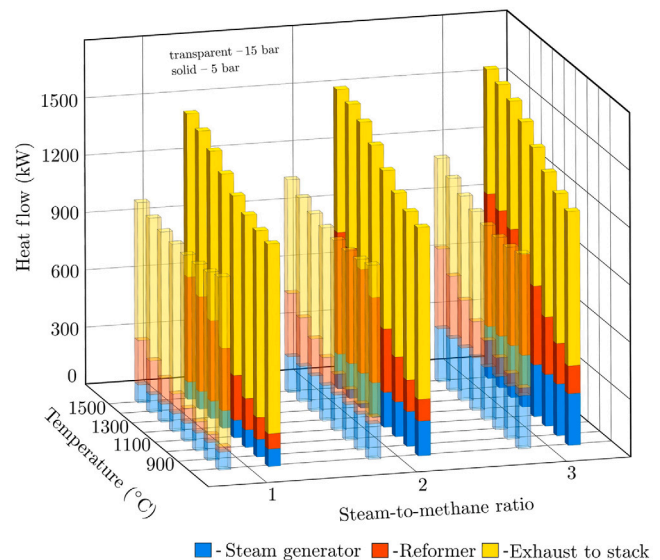


Fig. 8. Heat flow from exhaust gas after gas turbine for $\beta = 1, 2, 3$ and inlet pressure of 5 (solid) and 15 (transparent) bar.

flow is spent on the reforming process, steam generation and residual heat is supplied to stack.

The data in Fig. 8 allow us to deduce the following inference:

- an increase in β ratio leads to an increase in exhaust heat. This is mainly due to the fact that the steam fraction in the combustion products increases with increasing the mass flow rate of the exhaust gas. In this case, the heat capacity of steam is higher than the heat capacity of air, which it replaces according to the diagram in Fig. 2. At the same time, an increase in β ratio leads to an increase in the heat flow for steam generation. The same effect of steam addition can be seen in steam injected gas turbines (STIG) [43,44]. In addition, an increase in β ratio leads to a notable increase in heat flow in the reformer. This is due to the fact that with an increase in β ratio, the enthalpy of the reforming process increases, and with an increase in the amount of steam in

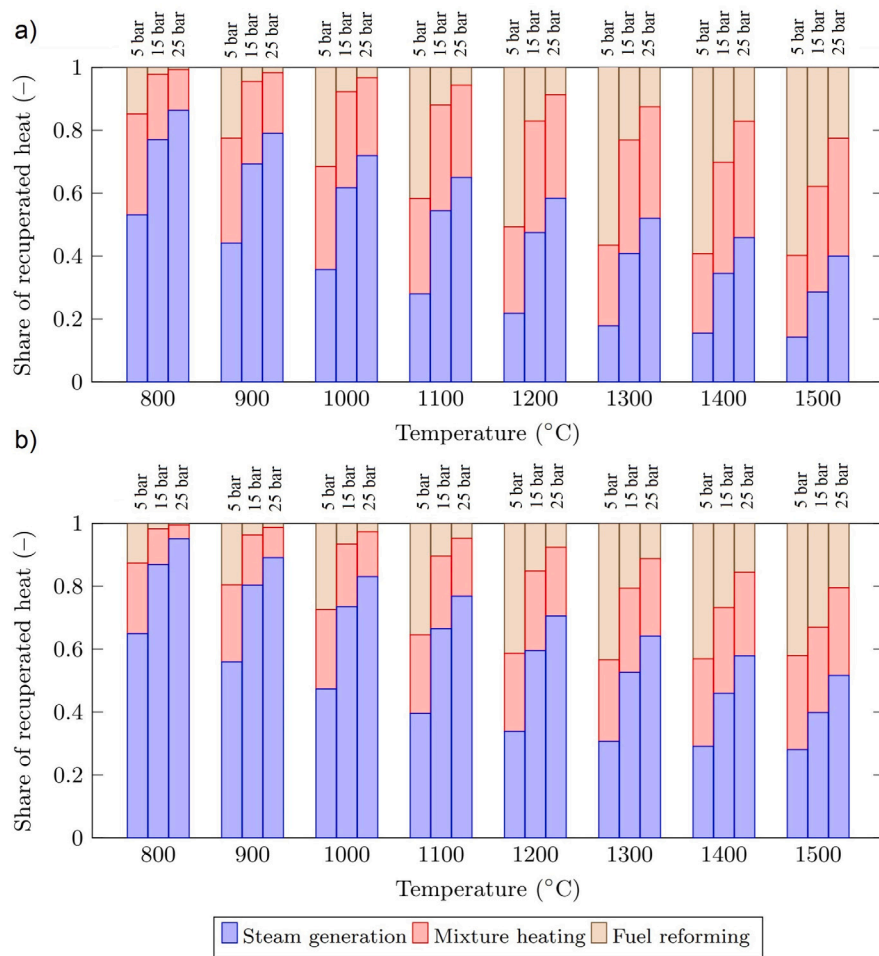


Fig. 9. Share of recuperated heat in each element of the thermochemical exhaust heat recuperation system for various operating parameters: (a) $\beta = 1$; (b) $\beta = 3$.

the initial steam-methane mixture, the heat flow for heating this mixture increases;

- an increase in pressure leads to a decrease in exhaust heat. This is a well-known fact [45]. However, an increase in pressure leads to a decrease in the recuperated heat in the reformer. This is due, firstly, to the fact that the temperature of the exhaust gases is lower and, as a consequence, the enthalpy of reforming is lower for a given gas temperature at the reformer inlet. Secondly, an increase in pressure leads to a decrease in the enthalpy of the process (Fig. 3);
- an increase in the gas temperature leads to an increase in the heat flow in the reformer because in this case the enthalpy of the process and the heat flow for heating the steam-methane mixture increase. At moderately low temperatures, the heat flow in the reformer is minimal. This is due to the fact that reforming practically does not take place in the reformer.

To understand how the recuperated heat is distributed in the thermochemical exhaust heat recuperation system, we estimated the share of recuperated heat in each element of the heat recuperation system according to the expression (5).

Fig. 9 shows the distribution of recuperated heat in a heat recuperation system versus temperature for different pressures and for $\beta = 1$ (Fig. 9a) and $\beta = 3$ (Fig. 9b). Fig. 9 shows that at low inlet gas temperatures, exhaust heat is recuperated mainly due to steam generation. The share of recuperated heat in the steam generator at $p = 25$ bar and at $T = 800$ °C is 0.98 and 0.97 for $\beta = 1$ and $\beta = 3$, respectively. As the temperature increases, the share of recuperated heat in the reformer increases. At $T = 1500$ °C and $p = 25$ bar, about 60% and 50% of the

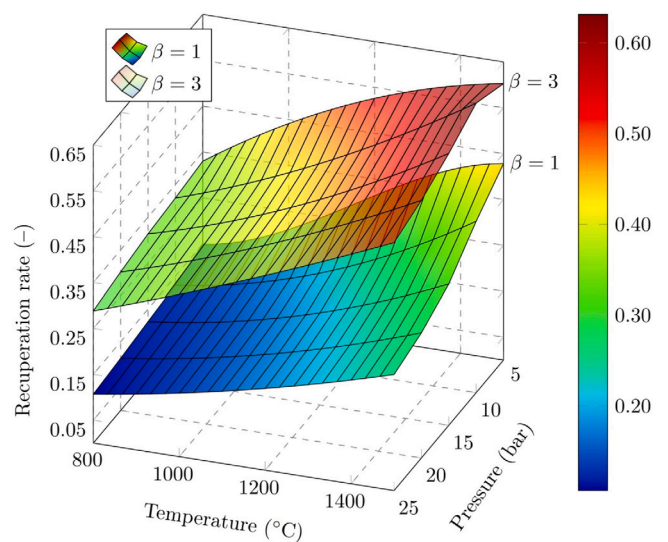


Fig. 10. The heat recuperation rate for the thermochemical exhaust heat recuperation system as a function of temperature (T_{in}) and inlet pressure for $\beta = 1$ and $\beta = 3$.

total recuperated heat is recuperated in the reformer, for $\beta = 1$ and $\beta = 3$, respectively. Increasing the steam-to-methane ratio (β) increases the share of recuperated heat in the steam generator. At $T = 1400$ °C for $\beta = 1$ and $p = 15$ bar, the share of recuperated heat due to steam

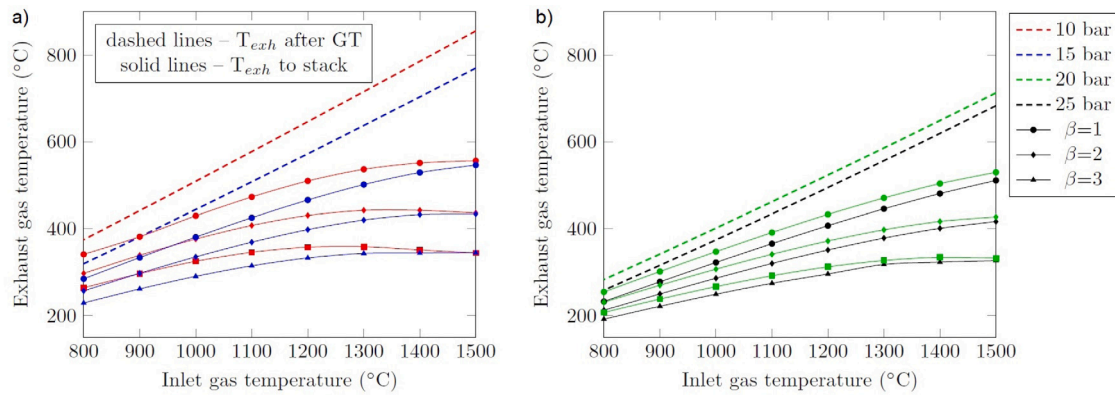


Fig. 11. Temperature of exhaust gas after a gas turbine (T_{ex}) and recuperation system as a function of inlet gas temperature for various pressure and steam-to-methane ratio: solid lines — temperature of exhaust gas after a recuperation system (to stack); dashed lines — temperature of exhaust gas after a gas turbine.

generation, mixture heating and methane reforming are approximately equal.

One of the key parameters for assessing the efficiency of the heat recuperation systems is the heat recuperation rate (Eq. (4)). Fig. 10 depicts the heat recuperation rate for the thermochemical exhaust heat recuperation system as a function of temperature and pressure for $\beta = 1$ and $\beta = 3$.

Fig. 10 shows that up to 63% of exhaust heat (for $(T_{in}) = 1500$ °C and $\beta = 3$) can be recuperated to the cycle of a gas turbine. Also, we can see that increase in temperature increases the heat recuperation rate. This is mainly due to an increase in the enthalpy of the reforming process when temperature increases. However, Fig. 10 shows that up to 85% (for $(T_{in}) = 800$ °C and $\beta = 1$) and up to 37% (for $(T_{in}) = 1500$ °C and $\beta = 3$) of exhaust heat is supplied to the stack. Therefore, the thermochemical exhaust heat recuperation systems can be combined with other recuperation ways, for example, the addition of steam cycle, air heating, etc.

To evaluate the potential of exhaust heat after the thermochemical exhaust heat recuperation system, temperatures of the exhaust gases to stack were determined. Fig. 11 shows temperature of exhaust gas after a gas turbine (T_{ex}) and after recuperation system as a function of inlet gas temperature for various pressure and steam-to-methane ratio. In this figure, the solid lines are temperatures of exhaust gas after a recuperation system (to stack), the dashed lines are temperatures of exhaust gas after a gas turbine. Colour of the lines is corresponding to each pressure: red — 10 bar, blue — 15 bar, green — 20 bar, black — 25 bar.

Fig. 11 shows the interesting findings. An increase in temperature of inlet gas the difference between gas temperature after a gas turbine and a recuperation system is increasing. Moreover, an increase in a steam-to-methane ratio also leads to an increase in the difference between gas temperatures before and after the recuperation system. For 10 bar, 15 bar and β ratio of 2, 3 we can see that an increase in temperature leads to decreasing in gas temperature after the recuperation system. That unusual behaviour can be explained by the fact that an increase in β ratio increases the enthalpy of the steam methane reforming process and, as a consequence, recuperated heat in the reformer. On the other hand, an increase in pressure above 20 bar has no noticeable effect on the gas temperature after the recuperation system. Gas temperature trends before and after the recuperation system are similar. In this regard, the following conclusion can be made that the use of the thermochemical exhaust heat recuperation system in GTU is the most effective in the temperature range of inlet gases from 1200 to 1500 °C at pressure of up to 25 bar.

5. Conclusion

The concept of a chemically recuperated gas turbine (CRGT) based on steam methane reforming was clarified in the details. To determine

the efficiency of CRGT as a function of operational parameters such as temperature, pressure, and steam-to-methane ratio before a reformer, the thermodynamic analysis was performed in Aspen HYSYS. The thermodynamic analysis of the gas turbine cycle and thermochemical recuperation system was performed for a wide range of operation parameters: temperature of gas at a reformer inlet of 800–1500 °C, expansion ratio of 5–25, inlet steam-to-methane ratio of 1–3.

The effect of inlet gas temperature on the efficiency of CRGT is general for gas turbines. An increase in the inlet gas temperature at the turbine inlet increases its efficiency. The efficiency of GTU with thermochemical recuperation can reach up to 0.48 (for $\beta = 3$) at an expansion ratio of 25. The increasing efficiency of CRGT due to thermochemical transformation is increasing with increasing temperature because an increase of temperature leads to an increase in methane conversion in the reformer.

The thermodynamic analysis of the thermochemical recuperation system showed that the distribution of recuperated heat in the elements of the recuperation system (reformer and steam generator) is strongly depending on operational parameters. The analysis showed that up to 63% of exhaust heat (for $T_{in} = 1500$ °C and $\beta = 3$) can be recuperated to the cycle of a gas turbine. An increase in temperature increases the heat recuperation rate.

Nomenclature

R	Heat recuperation rate;
H_i	Enthalpy of i th flow;
n_i	Number of moles of i th element;
η_{GTU}	Efficiency of GTU;
W_{gtu}	Power of gas turbine;
W_{comp}	Power of compressor;
m_{CH_4}	Mass flow rate of methane;
β	Steam-to-methane ratio (mol-to-mol);
μ_i	Chemical potential of element i ;
G^t	Value of total Gibbs free energy;
G_{fi}^0	Standard Gibbs free energy of formation of element i ;
R	Molar gas constant;
T_{in}	Gas temperature at the turbine inlet;
T_{ex}	Gas temperature at the turbine outlet;
f_i	Fugacity of pure elements i ;
f_i^0	Fugacity of pure elements i at the standard state;
C_{p_i}	Specific heat of i component;
LHV	Low heating value;
HRR	Heat recuperation rate;
TCR	Thermochemical recuperation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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